ALL CODES OUTPUTS OF DSA

Name: Mayank Kannamwar

PRN: B25ME1027

Batch: SY3-C3

EXPERIMENT 1

```
<<-----------Menu------------>>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 1
Enter String: mayank
Length of the String is 6

ENTER CHOICE (7 for Display Menu): 2
Enter String: Mayank
Enter second String: Maxxie
Original First String: Maxxie
Copied String: Maxxie

ENTER CHOICE (7 for Display Menu): 3
Enter String: mayank
Reversed String: knayam

ENTER CHOICE (7 for Display Menu): 4
Enter String: mmayankk
The String is NOT Palindrome.

ENTER CHOICE (7 for Display Menu): 5
Enter String: mayank
Enter substring String: may
The String B is substring of A and occurs 1 time(s).
```

```
ENTER CHOICE (7 for Display Menu): 6
Enter String: mayank
Enter second String: kannamwar
Strings are NOT Equal.

ENTER CHOICE (7 for Display Menu): 7
<<-----------Menu------------>>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 8


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 2

```
Enter the number of elements:3
Enter 3 elements
1 3 5

 Array elements are:
1 3 5

----MENU----
1. Linear Search
2. Binary Search
3. Display Array
4. Exit
Enter your choice: 1

Enter the elements to search: 2
Element is not found

----MENU----
1. Linear Search
2. Binary Search
3. Display Array
4. Exit
Enter your choice: 2
Enter the element to search: 3
Element found at position 2

----MENU----
1. Linear Search
2. Binary Search
3. Display Array
4. Exit
Enter your choice: 3
1 3 5
```

```
----MENU----
1. Linear Search
2. Binary Search
3. Display Array
4. Exit
Enter your choice: 4
Exiting Program...


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 3

```
Enter number of elements: 4
Enter elements:
1 3 5 6 4

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: Array elements are:
1 3 5 6

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 1
Array after Bubble Sort:
1 3 5 6

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 2
Array after Selection Sort:
1 3 5 6
```

```
--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 5
Exiting program...


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 4

```
====== STUDENT MENU ======
1. Create Student Details
2. Display Student Details
3. Linear Search
4. Sort by Marks
5. Exit

Enter your choice: 1

Enter details of Student 1
Enter Roll No: 20
Enter Name: Mayank
Enter Marks: 23

Enter details of Student 2
Enter Roll No: 21
Enter Name: Shivam
Enter Marks: 29

Enter details of Student 3
Enter Roll No: 22
Enter Name: Om
Enter Marks: 33

Enter details of Student 4
Enter Roll No: Bhuvan
Enter Name: Enter Marks: 21

Enter details of Student 5
Enter Roll No: 25
Enter Name: Radhey
Enter Marks: 25
```

```
====== STUDENT MENU ======
1. Create Student Details
2. Display Student Details
3. Linear Search
4. Sort by Marks
5. Exit

Enter your choice: 2

--- Student Details ---

Student 1
Roll No: 20
Name: Mayank
Marks: 23.00

Student 2
Roll No: 21
Name: Shivam
Marks: 29.00

Student 3
Roll No: 22
Name: Om
Marks: 33.00

Student 4
Roll No: 32766
Name: Bhuvan
Marks: 21.00

Student 5
Roll No: 25
```

```
Name: Radhey
Marks: 25.00


====== STUDENT MENU ======
1. Create Student Details
2. Display Student Details
3. Linear Search
4. Sort by Marks
5. Exit

Enter your choice: 3

Enter Roll No to search: 29

Student not found!


====== STUDENT MENU ======
1. Create Student Details
2. Display Student Details
3. Linear Search
4. Sort by Marks
5. Exit

Enter your choice: 3

Enter Roll No to search: 22

Student Found!
Roll No: 22
Name: Om
Marks: 33.00
```

```
====== STUDENT MENU ======
1. Create Student Details
2. Display Student Details
3. Linear Search
4. Sort by Marks
5. Exit

Enter your choice: 4

Students are sorted according to their marks


====== STUDENT MENU ======
1. Create Student Details
2. Display Student Details
3. Linear Search
4. Sort by Marks
5. Exit

Enter your choice: 5

Exiting program...

...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 5

```
        implrmrnt of stack
 Main menu
 pushn
2.pop
3.display
4.exit
 enter your choice:1
Enter the item to be pushed:3


Do you want to continue? (Y/N): y

 Main menu
 pushn
2.pop
3.display
4.exit
 enter your choice:2

 the popped element is 3

Do you want to continue? (Y/N): y

 Main menu
 pushn
2.pop
3.display
4.exit
 enter your choice:3

 the stackis empty

Do you want to continue? (Y/N): y
```

```
Do you want to continue? (Y/N): y

 Main menu
 pushn
2.pop
3.display
4.exit
 enter your choice:4

Exiting program...
Invalid choice!

Do you want to continue? (Y/N): n


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 6

```
Linked list:
30->20->10->NULL
Popped element = 30
After the pop, the new linked list:
20->10->NULL
Popped element = 20
After the pop, the new linked list is:
10->NULL


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 7

```
Enter a postfix expression :5

the value of the expression is : 5.000000


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 8

```
Inorder traversal: 20 30 40 50 60 70 80
Preorder traversal: 50 30 20 40 70 60 80
Postorder traversal: 20 40 30 60 80 70 50

Enter value to search: 20
Node 20 is present in the BST.


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 9

```
Edge    Weight
0 - 1   10
1 - 2   20
4 - 3   25
0 - 4   15
Total Weight of MST: 70


...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 10

